

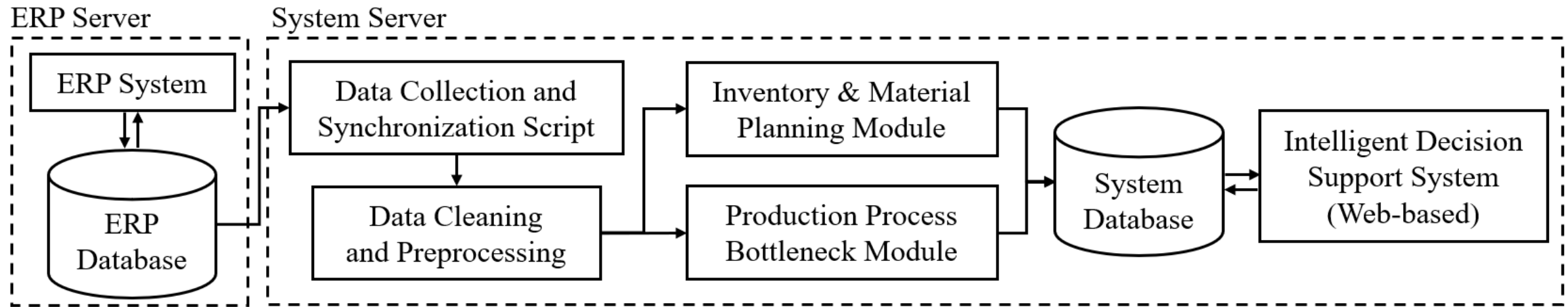
An Intelligent Decision Support System for Material Planning and Production Process Optimization in Smart Factories

Problem to Solve

- Inventory management
- Production bottlenecks
- Existing methods are difficult to deploy in real factories

How to Solve

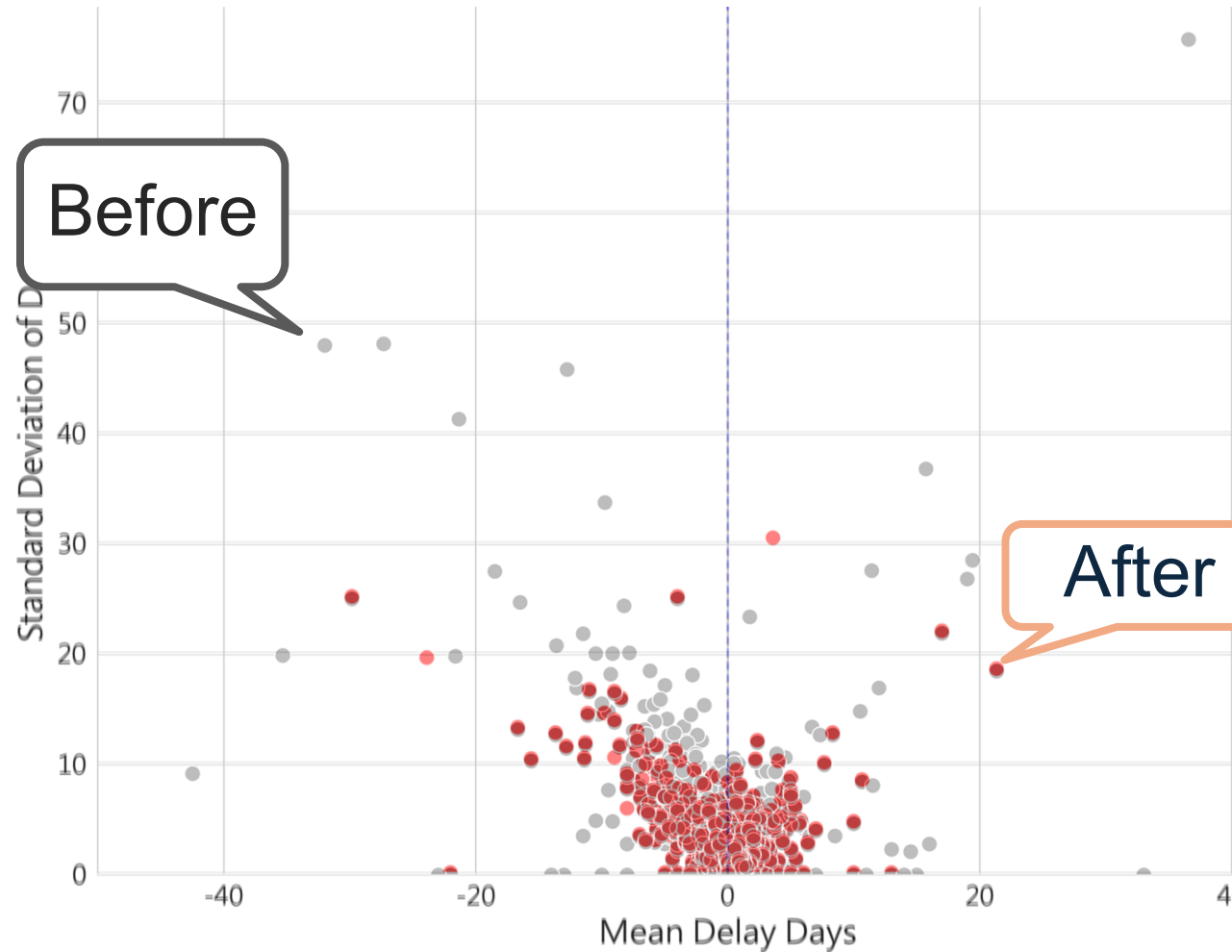
- Overall System Architecture



Data Source

ERP ID	Description	Content
MOCTA/B	Manufacturing Orders	Orders, processes, and required materials
INVMB	Material Master Data	Basic material information
PURTG	Inbound Records	Ordering and inbound information

Data Cleaning



- Removing missing values
- Retain production-related data
- Filter out low-frequency samples
- Exclude outliers using the $1.5 \times \text{IQR}$ rule

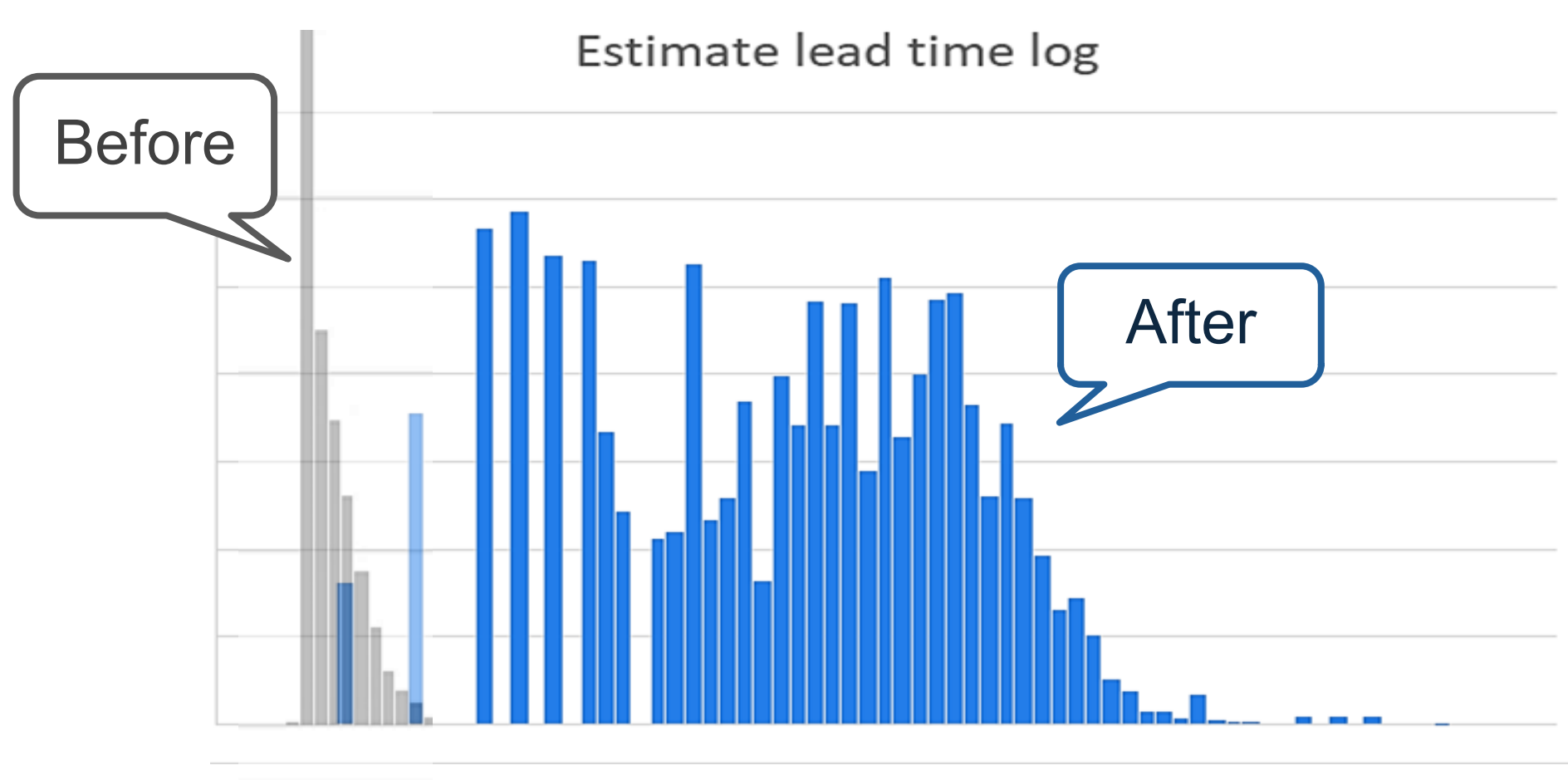
Two Predictive Models

- Material Lead-Time Prediction Model
- Process Completion Prediction Model

Feature Engineering & Preprocessing

- Static Features
 - ID, name, amount, specification, date...
- Dynamic Features
 - MA3, MA5, MAX, MIN, CV...
- Make Data Gaussian-like

Make Data Gaussian-like



Two Predictive Models' Result

Model	RMSE	MAE	MAPE (%)	R ²
LightGBM	2.550	1.394	10.20	0.984
XGBoost	2.528	1.396	10.26	0.985
CatBoost	2.616	1.494	10.86	0.983
NN	3.911	2.455	13.80	0.965

System Interface

Functions we want to achieve

- View manufacturing orders, processes, and materials
- Support data editing and synchronization
- Provide a practical to-do list for daily operations

System Interface – Main Page

Form1

MO No.

Proc. Name	Product Name	Material No	Material Name	Material Spec.	Delay Risk	Sched. Start	Sug. Start	Exp. Delay	Act. Start
MO No.: 1511-20241029001									
MO No.: 1511-20241029002									
MO No.: 1511-20241029003									
MO No.: 1511-20251103005									
MO No.: 1511-20260130001									
MO No.: 1511-20260130002									
MO No.: 1511-20260130003									
MO No.: 1511-20260130004									
MO No.: 1511-20260130005									
MO No.: 1511-20260306001									
MO No.: 1511-20260306002									
MO No.: 1512-20260211001									
09-Spindle Offline	Lower Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/25	2026/4/24	112	2026/2/25
09-Spindle Offline	Lower Long Shaft C ...	M0CA01ZZ000129	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/25	2026/4/24	112	2026/2/25
Precision Assembly	Lower Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...		2026/2/25			2026/2/25
Precision Assembly ...	Lower Long Shaft C ...	M0CA01ZZ000129	Drive Key Coupling	For 2nd Gen Long S...		2026/2/25			2026/2/25
22-System Module 3 S...	Lower Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/25	2026/11/22	120	2026/2/25
22-System Module 3 S...	Lower Long Shaft C ...	M0CA01ZZ000129	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/25	2026/11/22	120	2026/2/25
MO No.: 1512-20260211002									
09-Spindle Offline	Higher Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/23	2026/4/24	112	2026/2/23
09-Spindle Offline	Higher Long Shaft C ...	M0CA01ZZ000128	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/23	2026/4/24	112	2026/2/23
Precision Assembly ...	Higher Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...		2026/2/23			2026/2/23
Precision Assembly ...	Higher Long Shaft C ...	M0CA01ZZ000128	Drive Key Coupling	For 2nd Gen Long S...		2026/2/23			2026/2/23
22-System Module 3 S...	Higher Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/23	2026/11/22	120	2026/2/23
22-System Module 3 S...	Higher Long Shaft C ...	M0CA01ZZ000128	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/23	2026/11/22	120	2026/2/23
MO No.: 1512-20260211007									
MO No.: 1512-20260212002									
MO No.: 1512-20260312001									
09-Spindle Offline	Gear Shift Shaft Dyna...	M0113111310001	C-type Retaining Ring	(Consumables) for 10...	HIGH	2026/3/23	2026/4/24	112	2026/3/23
09-Spindle Offline	Gear Shift Shaft Dyna...	M0303030300004	Single Row Deep Gro...	6013	HIGH	2026/3/23	2026/4/24	112	2026/3/23
09-Spindle Offline	Gear Shift Shaft Dyna...	M0309030900019	Bearing Lock Nut	YSR M65*2.0/	HIGH	2026/3/23	2026/4/24	112	2026/3/23

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> MO No.: 1511-20241029002									
> MO No.: 1511-20241029003									
> MO No.: 1511-20251103005									
> MO No.: 1511-20260130001									
> MO No.: 1511-20260130002									
> MO No.: 1511-20260130003									
> MO No.: 1511-20260130004									
> MO No.: 1511-20260130005									
> MO No.: 1511-20260306001									
> MO No.: 1511-20260306002									
v MO No.: 1512-20260211001									
09-Spindle Offline	Lower Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/25	2026/4/24		112 2026/2/25
09-Spindle Offline	Lower Long Shaft C ...	M0CA01ZZ000129	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/25	2026/4/24		112 2026/2/25
Precision Assembly ...	Lower Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...		2026/2/25			2026/2/25
Precision Assembly ...	Lower Long Shaft C ...	M0CA01ZZ000129	Drive Key Coupling	For 2nd Gen Long S...		2026/2/25			2026/2/25
22-System Module 3 S...	Lower Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/25	2026/11/22		120 2026/2/25
22-System Module 3 S...	Lower Long Shaft C ...	M0CA01ZZ000129	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/25	2026/11/22		120 2026/2/25
v MO No.: 1512-20260211002									
09-Spindle Offline	Higher Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/23	2026/4/24		112 2026/2/23
09-Spindle Offline	Higher Long Shaft C ...	M0CA01ZZ000128	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/23	2026/4/24		112 2026/2/23
Precision Assembly ...	Higher Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...		2026/2/23			2026/2/23
Precision Assembly ...	Higher Long Shaft C ...	M0CA01ZZ000128	Drive Key Coupling	For 2nd Gen Long S...		2026/2/23			2026/2/23
22-System Module 3 S...	Higher Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/23	2026/11/22		120 2026/2/23
22-System Module 3 S...	Higher Long Shaft C ...	M0CA01ZZ000128	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/23	2026/11/22		120 2026/2/23
> MO No.: 1512-20260211007									
> MO No.: 1512-20260212002									
v MO No.: 1512-20260312001									
09-Spindle Offline	Gear Shift Shaft Dyna...	M0113111310001	C-type Retaining Ring	(Consumables) for 10...	HIGH	2026/3/23	2026/4/24		112 2026/3/23
09-Spindle Offline	Gear Shift Shaft Dyna...	M0303030300004	Single Row Deep Gro...	6013	HIGH	2026/3/23	2026/4/24		112 2026/3/23
09-Spindle Offline	Gear Shift Shaft Dyna...	M0309030900019	Bearing Lock Nut	YSR M65*2.0/	HIGH	2026/3/23	2026/4/24		112 2026/3/23

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> MO No.: 1511-20241029003									
> MO No.: 1511-20251103005									
> MO No.: 1511-20260130001									
> MO No.: 1511-20260130002									
> MO No.: 1511-20260130003									
> MO No.: 1511-20260130004									
> MO No.: 1511-20260130005									
> MO No.: 1511-20260306001									
> MO No.: 1511-20260306002									
v MO No.: 1512-20260211001									
09-Spindle Offline	Lower Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...	HIGH	2026/2/25	2026/4/24		112 2026/2/25
09-Spindle Offline	Lower Long Shaft C ...	M0CA01ZZ000129	Drive Key Coupling	For 2nd Gen Long S...	HIGH	2026/2/25	2026/4/24		112 2026/2/25
Precision Assembly ...	Lower Long Shaft C ...	M03230ZZ000007	Tapered Coupling	External Resistance ...		2026/2/25			2026/2/25
Precision Assembly ...	Lower Long Shaft C ...	M0CA01ZZ000129	Drive Key Coupling	For 2nd Gen Long S...		2026/2/25			2026/2/25
22-System Module 3 S...				External Resistance ...	HIGH	2026/2/25	2026/11/22		120 2026/2/25
22-System Module 3 S...				For 2nd Gen Long S...					120 2026/2/25
v MO No.: 1512-20260211001									
09-Spindle Offline				External Resistance ...					112 2026/2/23
09-Spindle Offline				For 2nd Gen Long S...					112 2026/2/23
Precision Assembly ...				External Resistance ...					2026/2/23
Precision Assembly ...				For 2nd Gen Long S...					2026/2/23
22-System Module 3 S...				External Resistance ...					120 2026/2/23
22-System Module 3 S...				For 2nd Gen Long S...					120 2026/2/23
> MO No.: 1512-20260211001									
> MO No.: 1512-20260211001									
v MO No.: 1512-20260211001									
09-Spindle Offline	Gear Shift Shaft Dyna...	M0113111310001	C-type Retaining Ring	(Consumables) for 10...	HIGH	2026/3/23	2026/4/24		112 2026/3/23
09-Spindle Offline	Gear Shift Shaft Dyna...	M0303030300004	Single Row Deep Gro...	6013	HIGH	2026/3/23	2026/4/24		112 2026/3/23
09-Spindle Offline	Gear Shift Shaft Dyna...	M0309030900019	Bearing Lock Nut	YSR M65*2.0/	HIGH	2026/3/23	2026/4/24		112 2026/3/23

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112 2026/3/23
112 2026/3/23

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System Interface – BOM Page

XtraForm1

MONo.

Material No	Material Name	Exp.Qty	ShortageQty	Act.Start	Exp.Material Delay Day	Exp.Delivery Date	Exp.Machine Delay Day	Risk Assessment
> MO No. : 1511-20241029001								
> MO No. : 1511-20241029002								
> MO No. : 1511-20241029003								
> MO No. : 1511-20260130001								
> MO No. : 1511-20260130002								
v MO No. : 1511-20260130003								
M0J961ZZ000035	Vertical Finished Machine	1		1	2026-04-24	1	2026-04-01	-23 LOW
> MO No. : 1511-20260130004								
v MO No. : 1511-20260130005								
M0J961ZZ000037	Vertical Finished Machine	1						-23 LOW
> MO No. : 1511-20260306001								
> MO No. : 1511-20260306002								
v MO No. : 1512-20260211001								
M03230ZZ000007	Tapered Coupling	1						75 HIGH
M03230ZZ000007	Tapered Coupling	1						75 HIGH
M03230ZZ000007	Tapered Coupling	1						75 HIGH
> MO No. : 1512-20260211002								
> MO No. : 1512-20260211007								
> MO No. : 1512-20260212002								
> MO No. : 1512-20260312001								
> MO No. : 1531-20260206002								

Filter Editor

And

Risk Assessment Is not null

Risk Assessment

OK Cancel Apply

x ☒ Risk Assessment Is not null And 風險評估 ≠ Edit Filter

System Interface – BOM Page

XtraForm1

MONo. ▲

Material No	MaterialName	Exp.Qty	ShortageQty	Act.Start	Exp.Material Delay Day	Exp.Delivery Date	Exp.Machine Delay Day	RiskAssessment
> MO No. : 1511-20241029001								
> MO No. : 1511-20241029002								
> MO No. : 1511-20241029003								
1		1		2026-04-24		2026-04-01	-23	LOW
1		1					-23	LOW
1		1					75	HIGH
1		1					75	HIGH
1		1					75	HIGH
> MO No. : 1512-20260211002								
> MO No. : 1512-20260211007								
> MO No. : 1512-20260212002								
> MO No. : 1512-20260312001								
> MO No. : 1531-20260206002								

Filter Editor

And

RiskAssessment Is not null

RiskAssessment

Filter Editor

And

RiskAssessment Is not null

RiskAssessment

OK Cancel Apply

✕ ☒ RiskAssessment Is not null And 風險評估 ☐ Edit Filter

conclusion

- Validated with real factory ERP data
- Integrated material planning and bottleneck analysis
- Translated model outputs into a deployable decision support system
- Supports more practical scheduling and material-planning decisions

THANK YOU